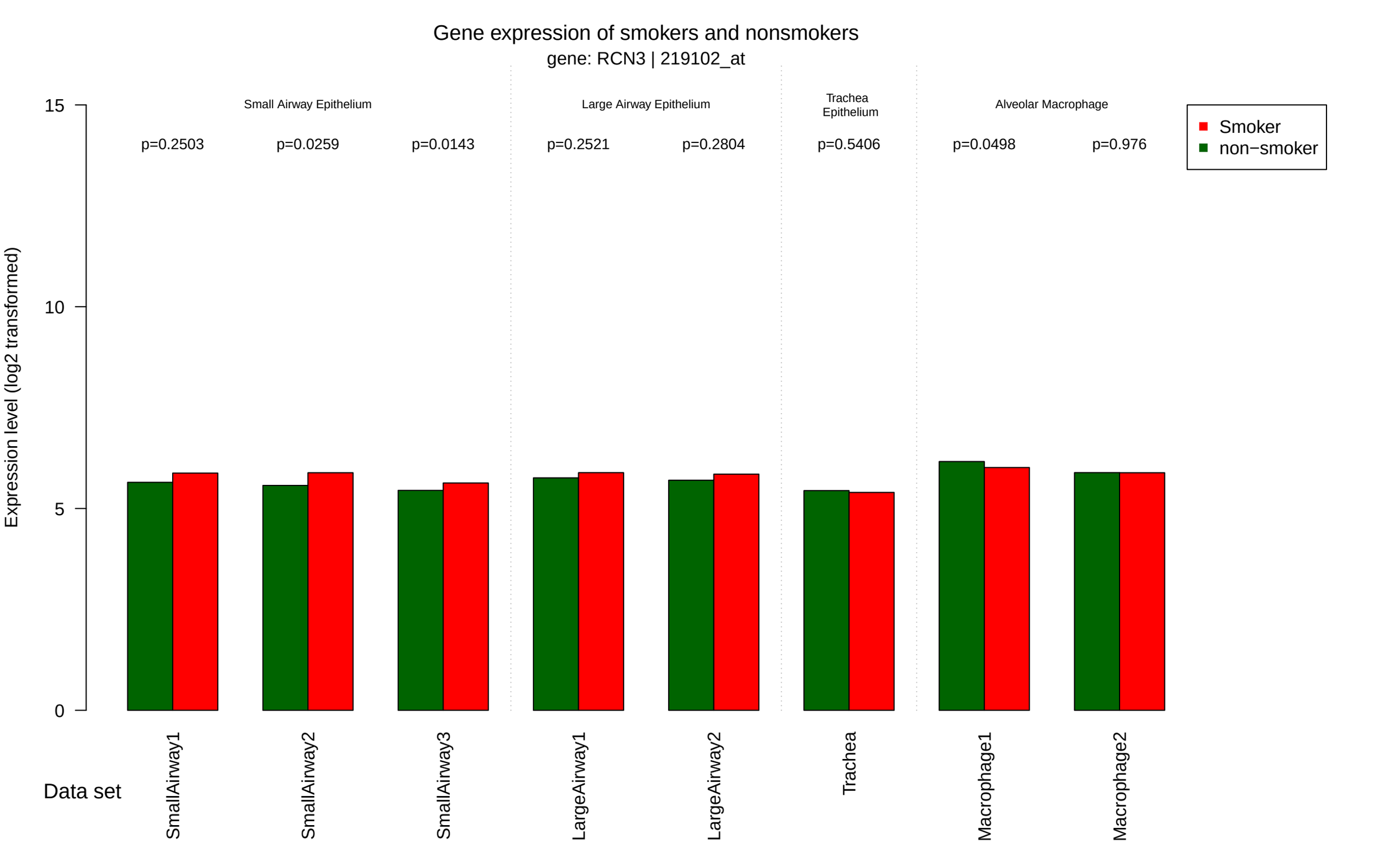
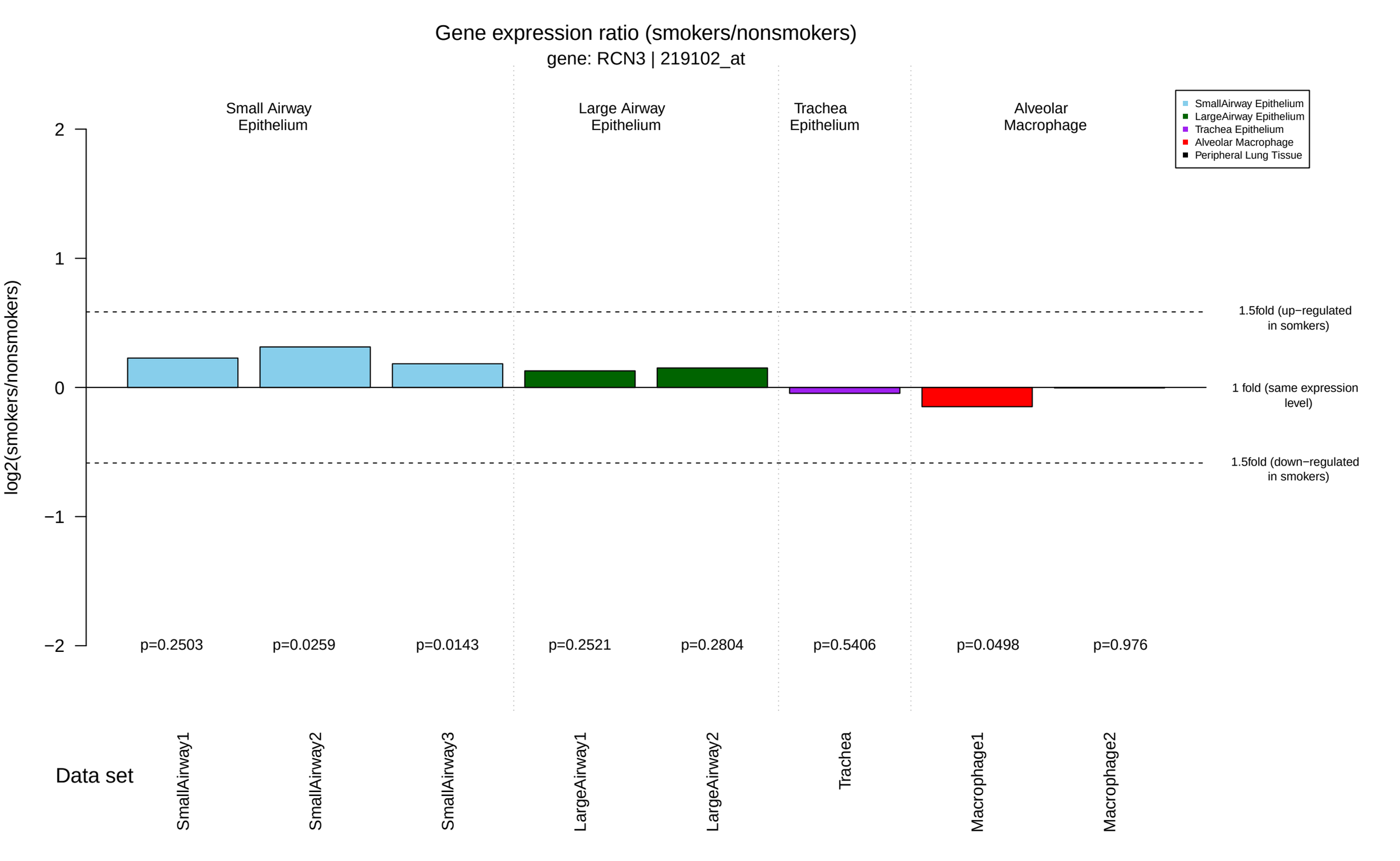
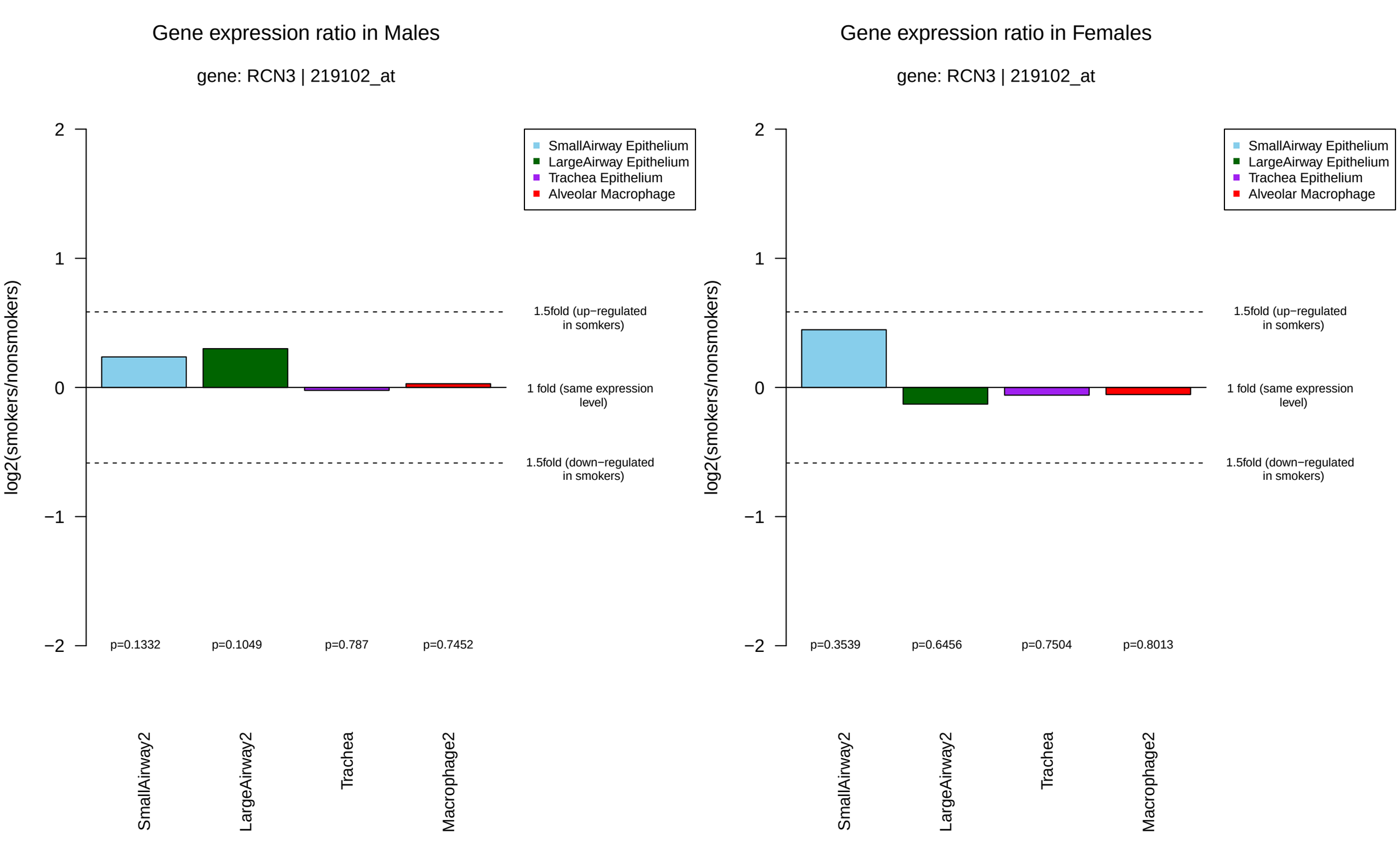
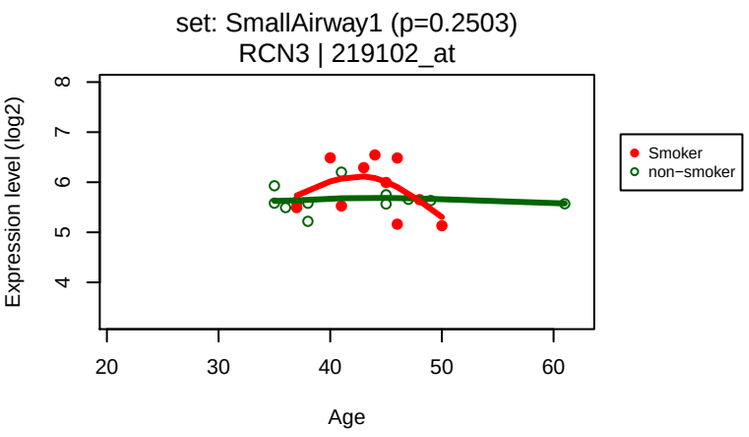
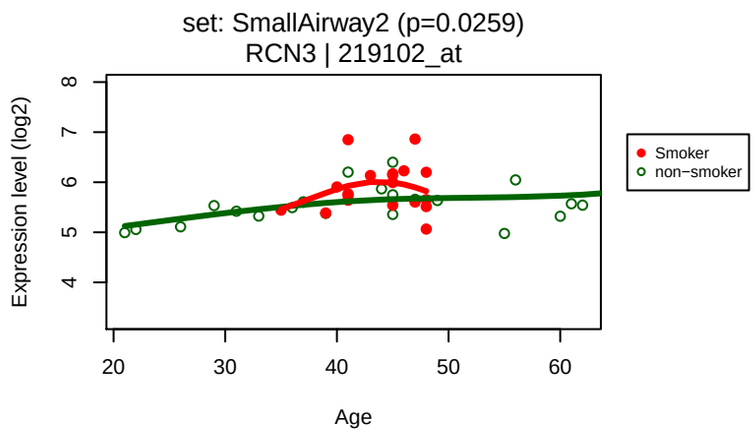
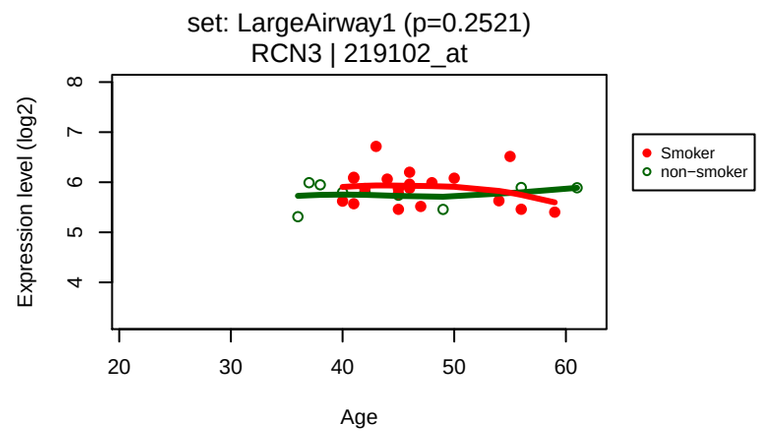

Gene expression of smokers and nonsmokers
gene: RCN3 | 219102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2503
p=0.0259
p=0.0143
p=0.2521
p=0.2804
p=0.5406
p=0.0498
p=0.976
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

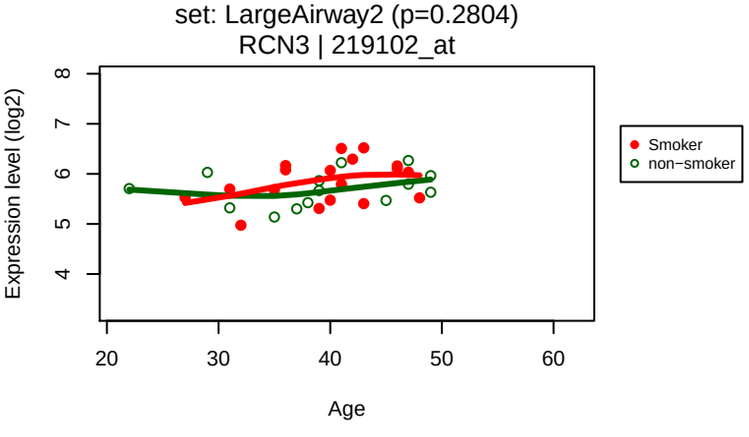

Gene expression ratio (smokers/nonsmokers)
gene: RCN3 | 219102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2503
p=0.0259
p=0.0143
p=0.2521
p=0.2804
p=0.5406
p=0.0498
p=0.976
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

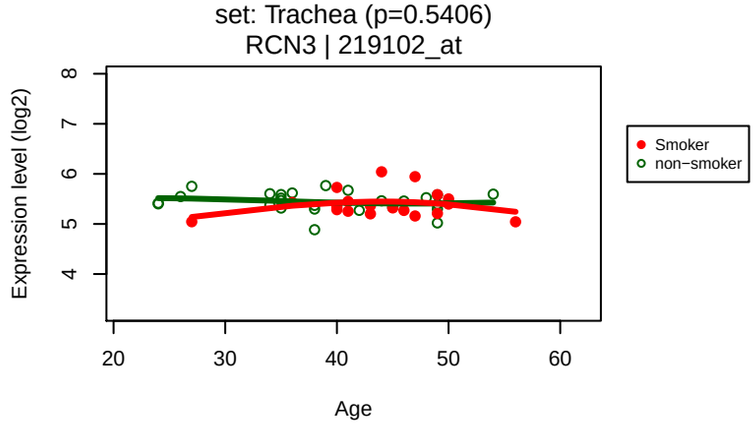

Gene expression ratio in Males
gene: RCN3 | 219102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1332
p=0.1049
p=0.787
p=0.7452
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCN3 | 219102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3539
p=0.6456
p=0.7504
p=0.8013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

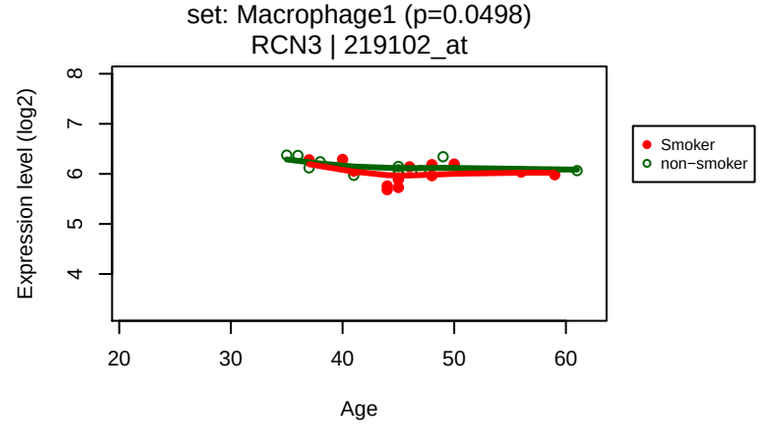

set: SmallAirway1 (p=0.2503)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker

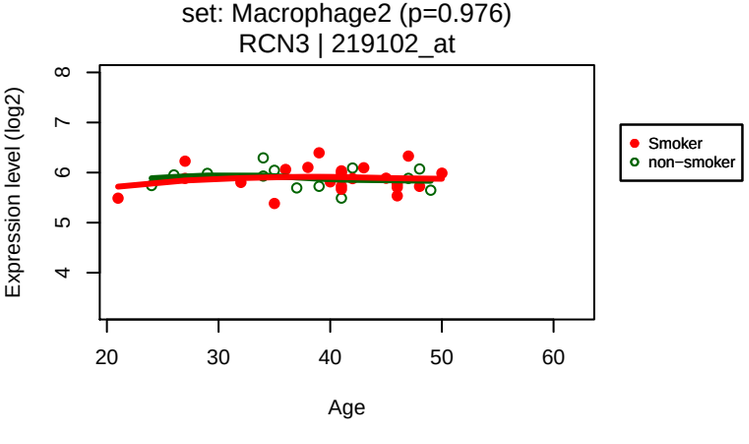

set: SmallAirway2 (p=0.0259)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker

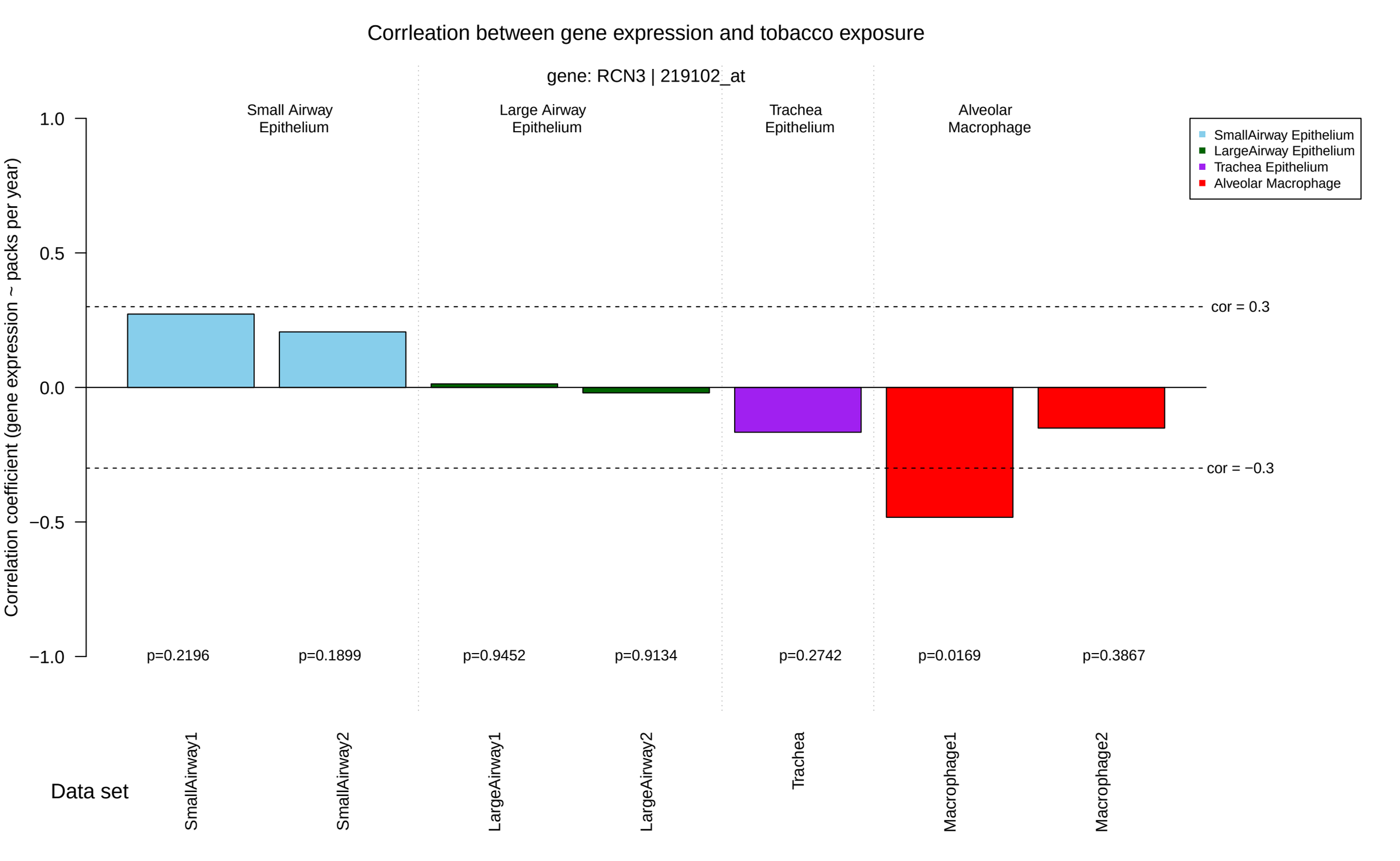

set: LargeAirway1 (p=0.2521)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2804)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5406)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0498)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.976)
RCN3 | 219102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCN3 | 219102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2196
p=0.1899
p=0.9452
p=0.9134
p=0.2742
p=0.0169
p=0.3867
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2